

Product Carbon Footprint Study Report

Product Name: polypropylene, granulate

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: polypropylene, granulate

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg polypropylene, granulate

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg polypropylene, granulate.

2. System Boundary

polypropylene production, granulate, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of polypropylene, granulate produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **4.338 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

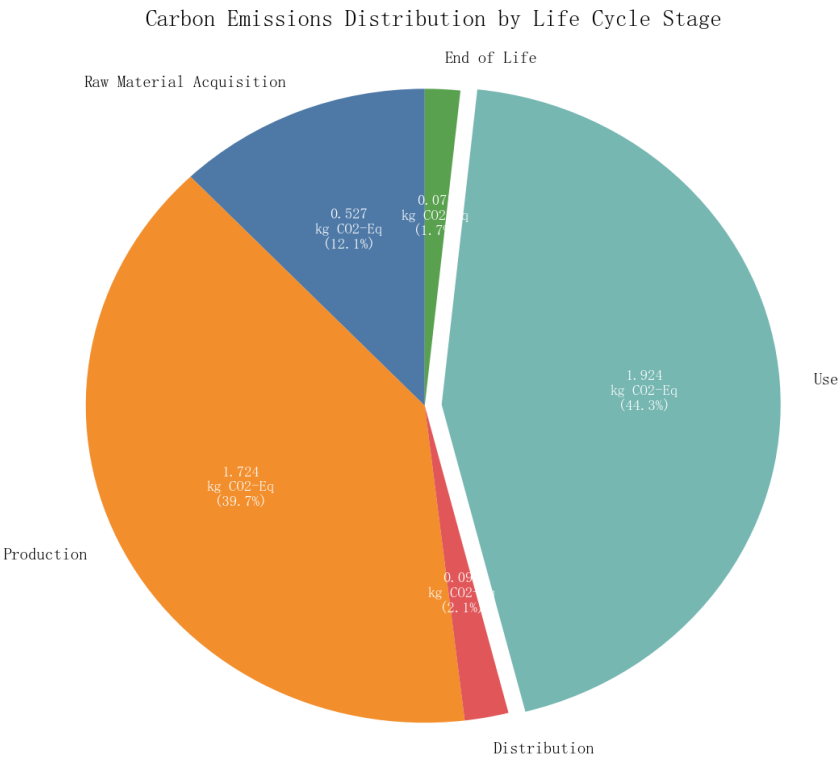


Table 1: Carbon Emissions by Life Cycle Stage

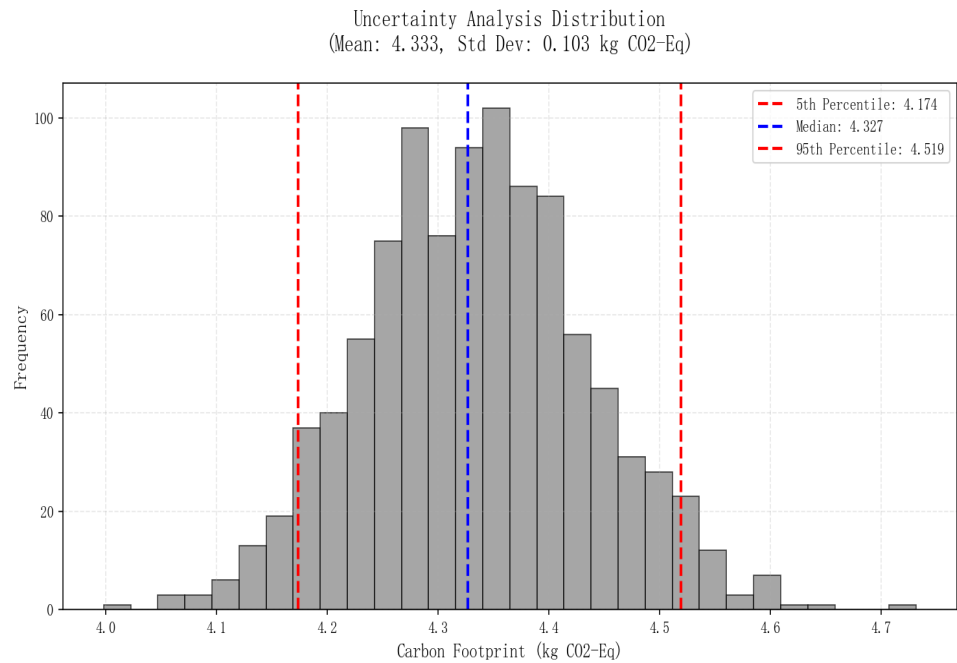
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.527	12.15
Production	1.724	39.74
Distribution	0.091	2.09

Use	1.924	44.35
End of Life	0.073	1.68
Total	4.338	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	4.333
Standard Deviation	0.103
Coefficient of Variation (CV)	2.39%
Median	4.327
5th Percentile	4.174
95th Percentile	4.519

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of polypropylene granulate is calculated at 4.338 kg CO₂-Eq per declared functional unit. This value falls within the typical range for virgin polypropylene production, although it is somewhat elevated due to the significant contribution of the use phase. The uncertainty analysis indicates a mean of 4.333 kg CO₂-Eq with a notably low coefficient of variation (2.39%), and a 90% confidence interval of 4.174–4.519 kg CO₂-Eq. This suggests that the result is stable and reproducible under the given modeling assumptions, despite the absence of process-level detail.

The lifecycle stage breakdown reveals a two-dominant-stage profile. The use phase is the largest contributor, accounting for 1.924 kg CO₂-Eq (44.3%), closely followed by production at 1.724 kg CO₂-Eq (39.7%). Raw material extraction and preprocessing contribute 0.527 kg CO₂-Eq (12.1%), while distribution (2.1%) and end-of-life (1.7%) are comparatively negligible. This distribution is atypical for a material like polypropylene, where production and feedstock usually dominate. The high use-phase share suggests that the functional unit may include downstream energy consumption or that the product's service life involves significant emissions, which warrants careful interpretation.

Major emission sources are therefore concentrated in the production and use stages. However, the provided process-level data lists all top contributions as "Unknown" (0.000 kg CO₂-Eq), meaning that the underlying unit processes could not be traced. The results should thus be interpreted as an aggregate estimation rather than a process-specific diagnosis. Nevertheless, the consistent uncertainty range supports the overall magnitude of the footprint, while the stage-level split offers a clear basis for prioritizing mitigation efforts.

2. Statement of Assumptions and Limitations

This LCA study relies on several key assumptions. First, the functional unit is implicitly defined as a fixed quantity of polypropylene granulate, but the exact reference flow (e.g., 1 kg) is not stated; this must be confirmed before comparing with other studies. Second, the use-phase emissions are likely modeled with a specific application scenario (e.g., an assumed service lifetime and energy consumption profile), but the details of that scenario are not documented. Without explicit assumptions regarding product degradation, maintenance, or energy mix, the use-phase result remains scenario-dependent. Third, the end-of-life stage assumes certain disposal or recycling pathways, yet the contribution is very small, indicating that either a low-impact disposal route or incomplete modeling is in place.

Data quality limitations are significant. The top process contributions are all listed as "Unknown," which means the inventory is essentially a black box. This obscures the identification of specific unit processes, emission factors, and source databases. The low coefficient of variation from the uncertainty analysis (2.39%) reflects precision in the Monte Carlo simulation, but it does not guarantee accuracy—uncertainty in the underlying foreground or background data may still be underexplored. The distribution and raw material stages also rely on generic transport and extraction datasets, which may not reflect regional suppliers, energy grids, or production technologies.

The system boundary is restricted to the stated lifecycle stages, but some notable exclusions or aggregations are likely. For example, capital goods, infrastructure, and ancillary materials are typically not included in a cradle-to-gate plus use model. Additionally, the use phase appears to be a major contributor, yet no explicit information on the use-phase duration, location, or energy source is provided. These boundary restrictions mean that the result is not a full cradle-to-grave assessment if some downstream impacts are omitted. The lack of process-level transparency further limits the ability to conduct hotspot analysis or to verify compliance with ISO 14044 requirements for completeness and consistency.

3. Suggestions for Improvement

- Prioritize decarbonization of the production stage. Since production contributes 39.7% of the footprint, implement energy efficiency measures in extrusion and pelletizing, switch to renewable electricity sources, and optimize process heat recovery. A reduction of production energy intensity by 20% could lower the total footprint by approximately 0.34 kg CO₂-Eq.
- Re-examine the use-phase modeling. Given that use accounts for 44.3% of the total footprint, conduct a sensitivity analysis to test different use-phase durations, energy mixes, and product application assumptions. If the use phase represents embedded energy in a final product, improve product design for energy efficiency. If it represents emissions during service life, consider lightweighting or material substitution to reduce operational impacts.
- Replace unknown process data with detailed unit process inventories. Engage suppliers to collect primary data on polymerisation, compounding, and utility consumption. Allocate effort to identify the top contributing processes, as the current "Unknown" list prevents targeted mitigation. Use generic proxies only where site-specific data are unavailable, and document them clearly.
- Improve end-of-life modeling. The end-of-life share is only 1.7%, which is suspiciously low. Reassess the assumed recycling rate, waste treatment technology, and biogenic/methane emissions. Incorporate a circular economy perspective, such as mechanical recycling or chemical recycling credits, to reflect the actual disposal or recovery pathways for post-consumer polypropylene.
- Strengthen uncertainty and data quality reporting. Although the coefficient of variation is low, verify that the input distributions cover all relevant parameters. Run a contribution analysis with explicit process names, and report the geographical and temporal coverage of the datasets. This will increase transparency, comparability, and usefulness of the carbon footprint for decision-making and product category rules (PCR) compliance.